Module - 2

Assignment.

* Permutation without Repetition.

1. How many different 4-digit numbers can be formed using 1, 2, 3, 4, 5, 6 without repetition?

Given Digits are 1, 2, 3, 4, 5, 6

We need to form 4-digit numbers without repeating any digit.

Formula := $^nP_r = \frac{n!}{(n-r)!}$

where, $n = 6$ and $r = 4$

$$\therefore {}^6P_4 = \frac{6!}{(6-4)!}$$

$$= \frac{6!}{2!} = \frac{6 \times 5 \times 4 \times 3 \times 2!}{2!}$$

$$= 6 \times 5 \times 4 \times 3$$

$$= 360$$

360 different 4-digit numbers can be formed without repetition

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2 From 8 students, in how many ways can 3 students be selected and arranged for the positions of President, secretary and Treasurer?

Given

8 students

3 students be selected and arranged.

Formula : ${}^{n}P_r = \frac{n!}{(n-r)!}$

where

$n = 8, \quad r = 3$

$\therefore {}^{8}P_3 = \frac{8!}{(8-3)!}$

$= \frac{8!}{5!} = \frac{8 \times 7 \times 6 \times 5!}{5!}$

$= 8 \times 7 \times 6$

$= 336$ ways

There are 336 ways

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3. A Company has 7 employees. In how many ways can it assign Manager, ~~The~~ Team Leader and Developer positions to different employees?

~~Here~~ ~~$n = 7$~~ , ~~$r = 3$~~

Formula : ${}^{n}P_r = \frac{n!}{(n-r)!}$

Here $n = 7$ & $r = 3$

$\therefore {}^{7}P_3 = \frac{7!}{(7-3)!}$

$= \frac{7!}{4!} = \frac{7 \times 6 \times 5 \times 4!}{4!}$

$= 7 \times 6 \times 5$

$= 210$

There are total 210 ways for Manager, Team Leader and Developer positions to ~~diff~~ be assign for employees.

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4. A Company has 6 servers and wants to assign three different roles: Web server, Database server and Backup server. Each role must be assigned to a different server. In how many ways can be roles be assigned?

Formula $= {}^{n}P_{r} = \frac{n!}{(n-r)!}$

Where $n = 6, \quad r = 3$

$\therefore {}^{6}P_{3} = \frac{6!}{(6-3)!}$

$= \frac{6!}{3!} = \frac{6 \times 5 \times 4 \times 3!}{3!}$

$= 6 \times 5 \times 4$

$= 120$

In 120 ways the roles can be assigned.

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5 A network engineer has 5 different routers available. Three routers need to be selected & arranged in specific order for a test network. How many possible arrangements are there?

Formula $= {}^{n}P_{r} = \frac{n!}{(n-r)!}$

Here $n = 5$ & $r = 3$

$$\therefore {}^{5}P_{3} = \frac{5!}{(5-3)!}$$

$$= \frac{5!}{2!} = \frac{5 \times 4 \times 3 \times 2!}{2!}$$

$$= 5 \times 4 \times 3$$

$$= 60 \text{ ways}$$

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* Permutation with repetition

1. How many 3-digit numbers can be formed using the digits 2, 3, 4, 5 and 6 if repetition is allowed?

Total digits = 5
Since repetition is allowed, each position has 5 choice.

Formula = n^r

$= 5^3 = 5 \times 5 \times 5$

$= 125$

Total 125 numbers can be formed from 2, 3, 4, 5 & 6 digits.

2. How many 4-digits PIN's can be made using digits 0 through 9 if numbers can repeat?

Total digits = 10 (0 to 9)

Each position has 10 choices, and 0 can be the first digit of a PIN.

$= 10^4$

$= 10 \times 10 \times 10 \times 10$

$= 10,000$ PINs.

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3. Find how many ways you can rearrange letters of the word "BANANA" all at a time

Total letters = 6

- A is repeated 3 times.
- N is repeated 2 times
- B is repeated 1 times

Formula : $\frac{n!}{p! \times q!} = \frac{6!}{3! \times 2!} = \frac{720}{6 \times 2}$

$= \frac{720}{12} = 60$ arrangements.

4. Find the number of arrangements of the word "BALLOON".

Total letters = 7.

- L is repeated 2 times
- O is repeated 2 times
- B, A & N appear once each.

Formula : $\frac{n!}{p! q!} = \frac{7!}{2! \times 2!} = \frac{7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{2 \times 1 \times 2 \times 1}$

$= \frac{7 \times 6 \times 5 \times 4 \times 3}{2 \times 1}$

$= 1,260$ arrangements

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5 Find the number of arrangements of the word "APPLE".

Total letter = 5

- P is repeated 2 times.
- A, L and E appear once each

Formula : $\frac{n!}{p!}$

$= \frac{5!}{2!} = \frac{5 \times 4 \times 3 \times 2!}{2!}$

$= 60$ arrangements.

ha apple vala question optional ahe
~~nahi lihala tari chale lihala tari chalel~~
option tumch notes amchya

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5 Find the number of arrangements of the word "APPLE".

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$= 60$ arrangements.

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* Combination without repetition

1. A class has 5 students. We need to select a team of 2 students for a project.

- $n = 5, r = 2$

$$^5C_2 = \frac{5!}{2!(5-2)!}$$

$$= \frac{5!}{2! \cdot 3!} = \frac{120}{12}$$

$$= 10 \text{ ways}$$

2. Find the number of ways to select 3 books from 5 different books.

- $n = 5, r = 3$

$$^5C_3 = \frac{5!}{3!2!} = \frac{120}{12}$$

$$= 10 \text{ ways.}$$

3.

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3 Rahul chooses 5 marbles from 12 different colored marbles. In how many ways can she choose them?

$n = 12$, $r = 5$

$$^{12}C_5 = \frac{12!}{5!\,7!}$$

$$= \frac{12 \times 11 \times 10 \times 9 \times 8 \times 7!}{5! \times 7!}$$

$$= \frac{12 \times 11 \times 2 \times 3 \times 8}{5 \times 4 \times 3 \times 2 \times 1}$$

$$= 792$$

4 Jayesh's daughter chooses 4 pens from 18 different pens. How many ways can she do it?

$n = 18$, $r = 4$

$$^{18}C_4 = \frac{18!}{4! \times 14!}$$

$$= \frac{18 \times 17 \times 16 \times 15 \times 14!}{4! \times 14!}$$

$$= \frac{3 \quad 4}{18 \times 17 \times 16 \times 15}$$

$$= 3060 \text{ ways}$$

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5 You want to pick 2 different pizza toppings from 5 choices. In how many ways can it be done?

- $n = 5, \quad r = 2$

$$^5C_2 = \frac{5!}{2! \times 3!} = \frac{120}{12}$$

$$= 10 \text{ ways.}$$

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* Combination with Repetition

1. A shop has 3 types of fruits: Apple, Banana and Orange. Select 4 fruits, allowing repetition.

- $n = 3, r = 4$

Formula : ${}^{n+r-1}C_r = \frac{(n+r-1)!}{r!(n-1)!}$

$= {}^{3+4-1}C_4 = {}^{6}C_4$

$= \frac{6!}{4! \times 2!}$

$= 3\frac{6 \times 5 \times 4!}{4! \times 2!}$

$= 15$ ways

2. There are 4 flavours of ice cream. A customer wants 3 scoops, allowing repetition.

- $n = 4, r = 3$

$= {}^{4+3-1}C_3 = {}^{6}C_3$

$= \frac{6!}{3! \times 3!} = \frac{6 \times 5 \times 4 \times 3!}{3! \times 3!}$

$= \frac{6 \times 5 \times 4}{3 \times 2!}$

$= 20$ ways

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3. A bit string contains 0s & 1s. How many ways can we select 5 bits if repetition is allowed and order is ignored?

$n = 2, \quad r = 5$

$= {}^{2+5-1}C_5 = {}^{6}C_5$

$= \frac{6!}{5 \times 1!} = 6$ ways

4. You order 4 scoops of ice cream from 5 flavours, allowing repetition.

$n = 5, \quad r = 4$

$= {}^{5+4-1}C_4 = {}^{8}C_4$

$= \frac{8!}{4! \times 4!} = \frac{8 \times 7 \times 6 \times 5 \times 4!}{4! \times 4!}$

$= \frac{8 \times 7 \times 8 \times 5}{4 \times 3 \times 2 \times 1}$

$= 70$ Combinations

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5. A bakery sells 3 types of donuts. In how many ways can you choose a box of 6 dounts, allowing repetition?

$n = 3, \quad r = 6$

$= {}^{3+6-1}C_6 = {}^{8}C_6$

$= \frac{8!}{6! \times 2!} = \frac{8 \times 7 \times 6!}{6! \times 2!}$

$= \frac{8 \times 7}{2!}$

$= 28$ ways

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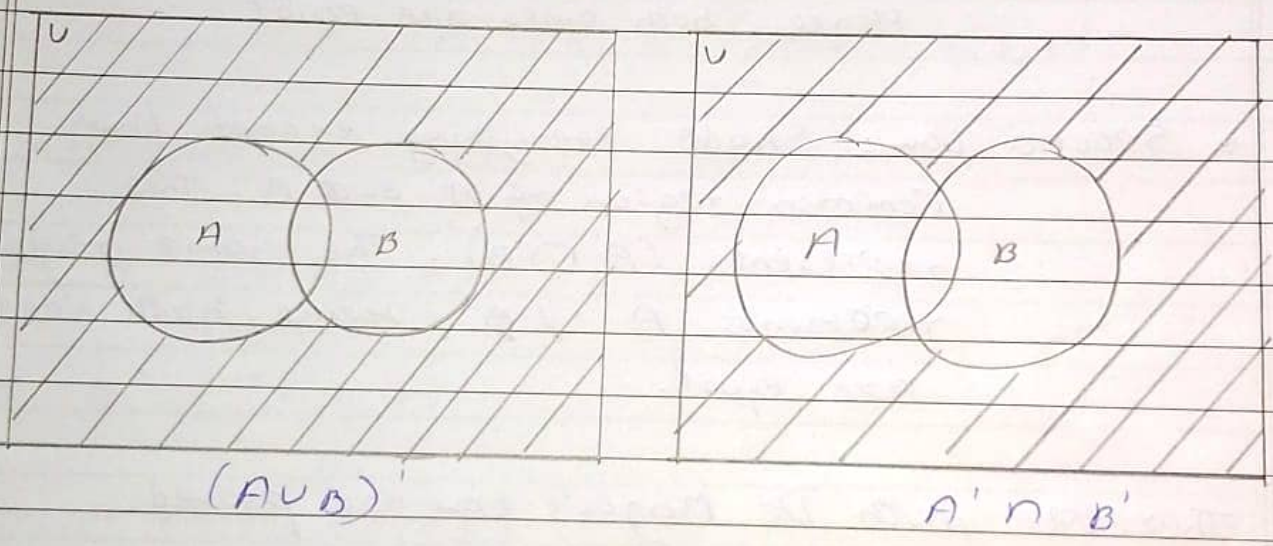
* De Morgan's Laws

1. State and prove De Morgan's Law of sets using Venn diagram.

De Morgan's Laws:

1. The complement of the union of two sets is equal to the intersection of their complements

~~$(A \cup B')'$~~ $(A \cup B)' = A' \cap B'$



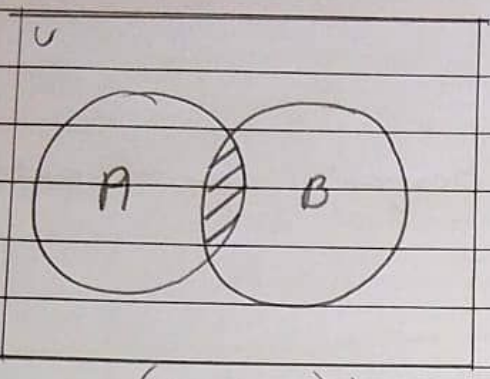
$(A \cup B)'$ $A' \cap B'$

2. The complement of the intersection of two sets is equal to the union of their complements.

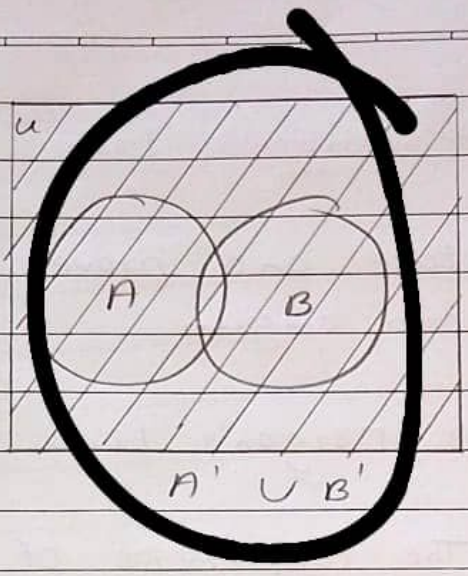
~~$A \cup$~~ $(A \cap B)' = A' \cup B'$

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$(A \cap B)'$



$A' \cup B'$

- First Law : Shade the region outside both sets A and B. This represents $(A \cup B)'$. The same region represents $A' \cap B'$. Hence, both sides are equal.

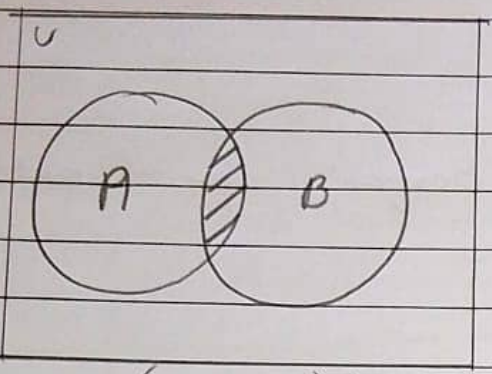
- Second Law : Shade everything except the common region of A and B. This represents $(A \cap B)'$. The same region represents $A' \cup B'$. Hence, both sides are equal.

Therefore, both De Morgan's Law are proved.

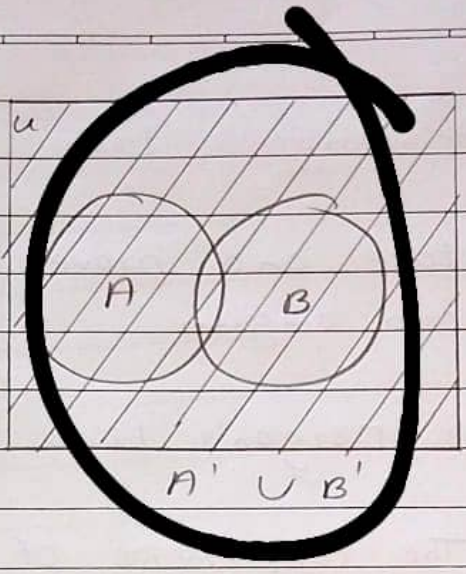
circle keleli diagram ahe tithe
blank space theva ti nantar
diagram pathavnar ahe tevha draw kara

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$(A \cap B)'$



$A' \cup B'$

- First Law: Shade the region outside both sets A and B. This represents $(A \cup B)'$. The same region represents $A' \cap B'$. Hence, both sides are equal.

- Second Law: Shade everything except the common region of A and B. This represents $(A \cap B)'$. The same region represents $A' \cup B'$. Hence, both sides are equal.

Therefore, both De Morgan's Law are proved.

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2 Let the Universal set $U = \{1, 2, 3, 4, 5, 6, 7\}$, $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$.

Find.

a. $A \cup B$

b. $(A \cup B)'$

c. A'

d. B'

e. $A' \cap B'$

- a. $A \cup B$ = $A = \{1, 2, 3, 4\}$
  $B = \{3, 4, 5, 6\}$
  $A \cup B = \{1, 2, 3, 4, 5, 6\}$

b. $(A \cup B)' = \{7\}$

c. $A' = \{5, 6, 7\}$

d. $B' = \{1, 2, 7\}$

e. $A' \cap B' = \{7\}$

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3 Let $U = \{a, b, c, d, e, f\}$, $A = \{a, b, c\}$ & $B = \{b, c, d\}$.
Find

a. $(A \cup B)'$

b. $(A \cap B)'$

c. $A' \cap B'$

d. $A' \cup B'$

a. $(A \cup B)' = A = \{a, b, c\}$

$B = \{b, c, d\}$

$(A \cup B) = \{a, b, c, d\}$

$(A \cup B)' = \{e, f\}$

b. $(A \cap B)' = (A \cap B) = \{b, c\}$

$(A \cap B)' = \{a, d, e, f\}$

c. $A' \cap B' = A' = \{d, e, f\}$, $B' = \{a, e, f\}$

$A' \cap B' = \{e, f\}$

d. $A' \cup B' = \{a, d, e, f\}$

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4 Let $U = \{0,1,2,3,4,5\}$, $A = \{0,1,3\}$ & $B = \{1,2,4\}$. Find.

a. $(A \cup B)'$
b. $(A \cap B)'$
c. $A' \cap B'$
d. $A' \cup B'$

a. $(A \cup B)' = (A \cup B) = \{0,1,2,3,4\}$
$(A \cup B)' = \{5\}$

b. $(A \cap B)' = A \cap B = \{1\}$
$(A \cap B)' = \{0,2,3,4,5\}$

c. $A' \cap B'$ $A' = \{2,4,5\}$ & $B' = \{0,3,5\}$
$A' \cap B' = \{5\}$

d. $A' \cup B' = \{0,2,3,4,5\}$

A

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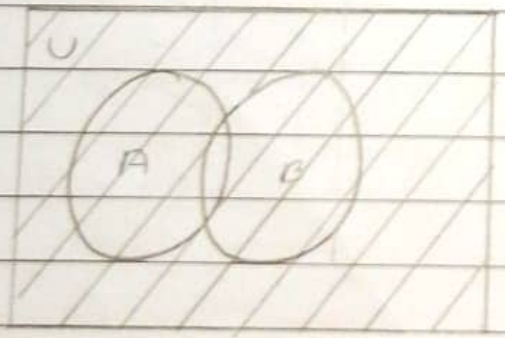
5 In a group of 50 people, A represents people who like coffee & B represents people who like tea. Express the set of people who do not like both coffee and tea using De Morgan's Law and represent it by a shaded region using venn diagram.

Given, . Universal set U = all 50 people
. A = people who like coffee
. B = people who like tea

We need to find: The set of people who do not like both coffee & tea.

- Express using De Morgan's Law
"Not both coffee and tea" means the complement of the intersection of A & B i.e. $(A \cap B)' = A' \cup B'$

- Venn diagram



Shade the region corresponding $A' \cup B'$. This includes
- The part outside both circles (neither coffee nor tea),
- The part only in A (coffee only)
- The part only in B (tea only)

The set of people who do not like both coffee and tea is $(A \cap B)' = A' \cup B'$

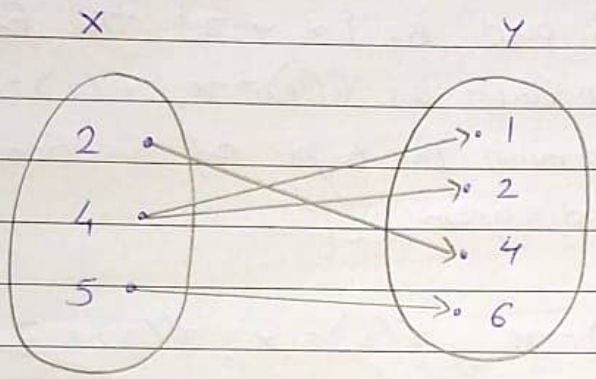
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* Function and its types.

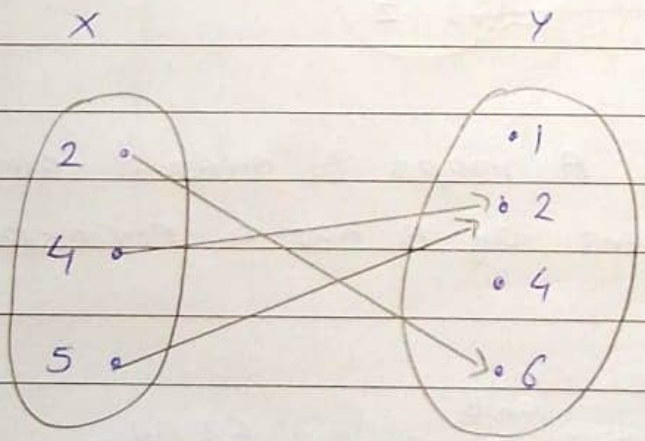
1. Let $X = \{2, 4, 5\}$ and $Y = \{1, 2, 4, 6\}$. Determine which of the following arrow diagrams determine fun^c from X to Y? Explain with reason with diagram does not represent a fun^c.

a.



- In this diagram, the element 4 has two arrows pointing 1 and 2
- It is not a function, because every element of X must have exactly one image in Y, Here the element 4 has two images.

b.



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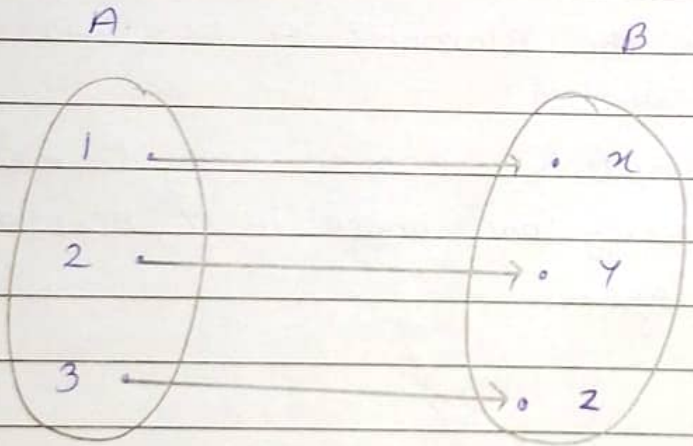
In diagram (b), the arrow are:

$2 \to 6$, $4 \to 2$, $5 \to 2$.

- Every element of X has exactly one image in Y.
- It is a many-to-one fun^c, because the elements 4 & 5 map to the same element 2 in Y.

2. Let $A = \{1, 2, 3\}$ and $B = \{x, y, z\}$. The function $f: A \to B$ is defined as $f(1) = x$, $f(2) = y$ and $f(3) = z$. Determine the type of function by using and arrow diagram.

Given : $f(1) = x$, $f(2) = y$, $f(3) = z$



Every element of A maps to different element of B, and every element of B has a corresponding element in A.

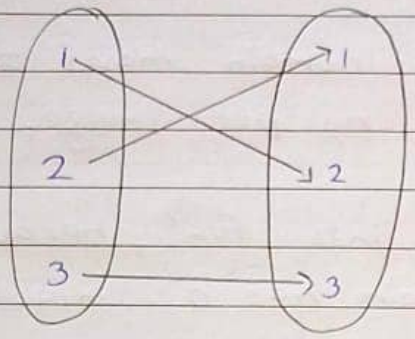
It s a one-to-one ~~or~~ and onto function

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3. Determine the type of function represented by an arrow diagram with reason.

a.



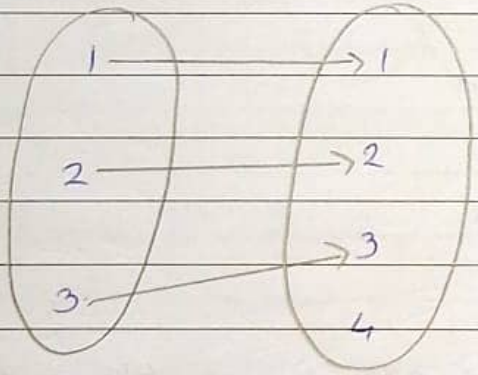
From the diagram

$f(1) = 2$, $f(2) = 1$, $f(3) = 3$

Each element of the domain maps to a different element of the co-domain and every element of the co-domain has a pre-image.

It is a one-to-one and onto (bijective) fun^c

b.



From the diagram

$f(1) = 1$, $f(2) = 2$, $f(3) = 3$

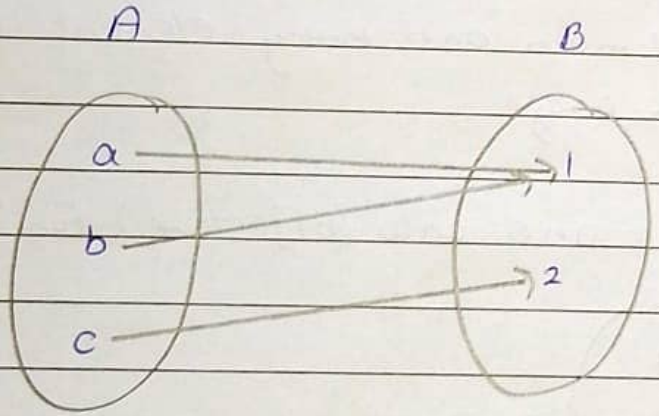
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The domain is $A = \{1, 2, 3\}$ and the co-domain is $B = \{1, 2, 3, 4\}$

Every element of the domain maps to a different element, but 4 has no pre-image.

It is a one-to-one into fun^c, because each element of the domain has a unique image, but range is not equal to the co-domain.

4. Determine the Domain, Co-domain and Range of the functions represented by following arrow diagram.



set $A = \{a, b, c\}$

set $B = \{1, 2\}$

1. Domain : The set of all input elements is called the domain

Domain $= \{a, b, c\}$

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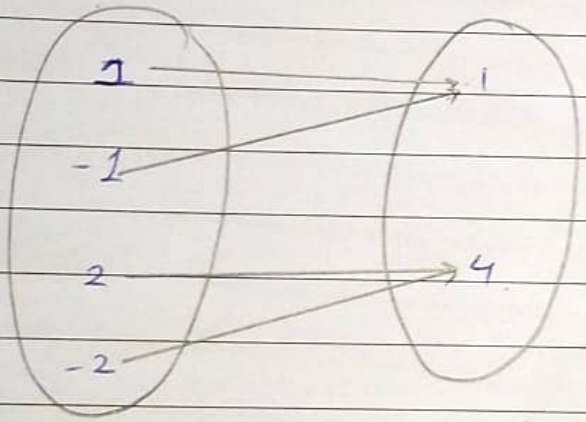
2. Co-domain ; The set containing all possible output elements is called the Co-domain.

Co-domain = {1, 2}

3. Range ; The set of output element that receive at ~~the~~ least one arrow is called the range.

Range = {1, 2}

5. Determine the Domain, Co-domain and Range of the fun' represented by following arrow diagram.



Set A = {1, -1, 2, -2}

Set B = {1, 4}

1. Domain : The set of all input elements is called the domain.

Domain = {1, -1, 2, -2}

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2. Co-domain : The set containing all possible output element is called the Co-domain

Co-domain = {1, 4}

3. Range : The set of output elements that receive at least one arrow is called the range.

Range = {1, 4}

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